FastAPI Learn Advanced User Guide

Sub Applications - Mounts

If you need to have two independent FastAPI applications, with their own independent OpenAPI and their own docs UIs, you can have a main app and "mount" one (or more) sub-application(s).

Mounting a FastAPI application

"Mounting" means adding a completely "independent" application in a specific path, that then takes care of handling everything under that path, with the *path operations* declared in that sub-application.

Top-level application

First, create the main, top-level, **FastAPI** application, and its *path operations*:

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/app")
def read_main():
    return {"message": "Hello World from main app"}


subapi = FastAPI()


@subapi.get("/sub")
def read_sub():
    return {"message": "Hello World from sub API"}


app.mount("/subapi", subapi)
```

Sub-application

Then, create your sub-application, and its *path operations*.

This sub-application is just another standard FastAPI application, but this is the one that will be "mounted":

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/app")
def read_main():
    return {"message": "Hello World from main app"}


subapi = FastAPI()
```

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```
@subapi.get("/sub")
def read_sub():
    return {"message": "Hello World from sub API"}


app.mount("/subapi", subapi)
```

Mount the sub-application

In your top-level application, `app`, mount the sub-application, `subapi`.

In this case, it will be mounted at the path `/subapi`:

Python 3.10+

```
from fastapi import FastAPI

app = FastAPI()


@app.get("/app")
def read_main():
    return {"message": "Hello World from main app"}


subapi = FastAPI()


@subapi.get("/sub")
def read_sub():
    return {"message": "Hello World from sub API"}


app.mount("/subapi", subapi)
```

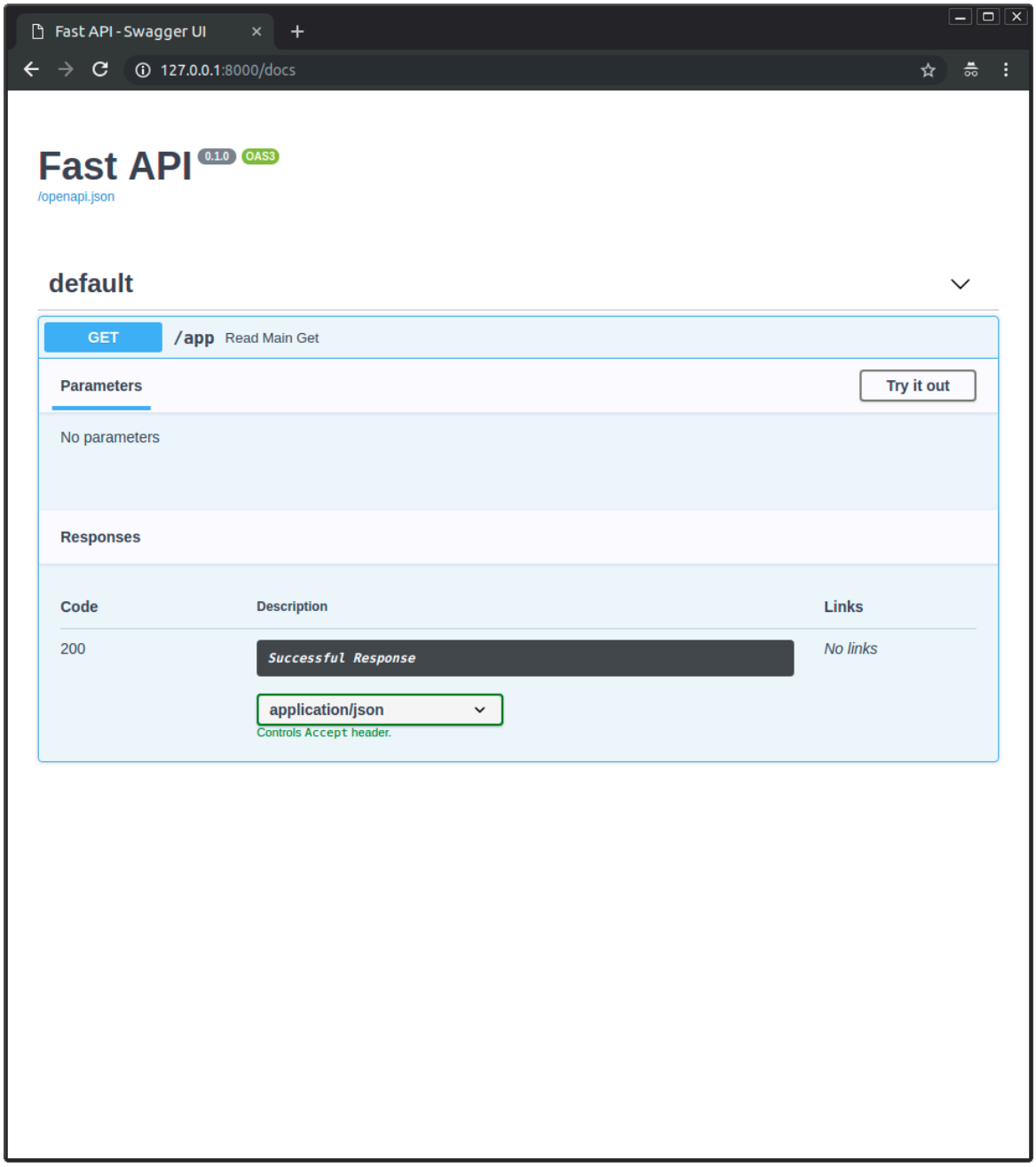
Check the automatic API docs

Now, run the `fastapi` command:

```
bash
```

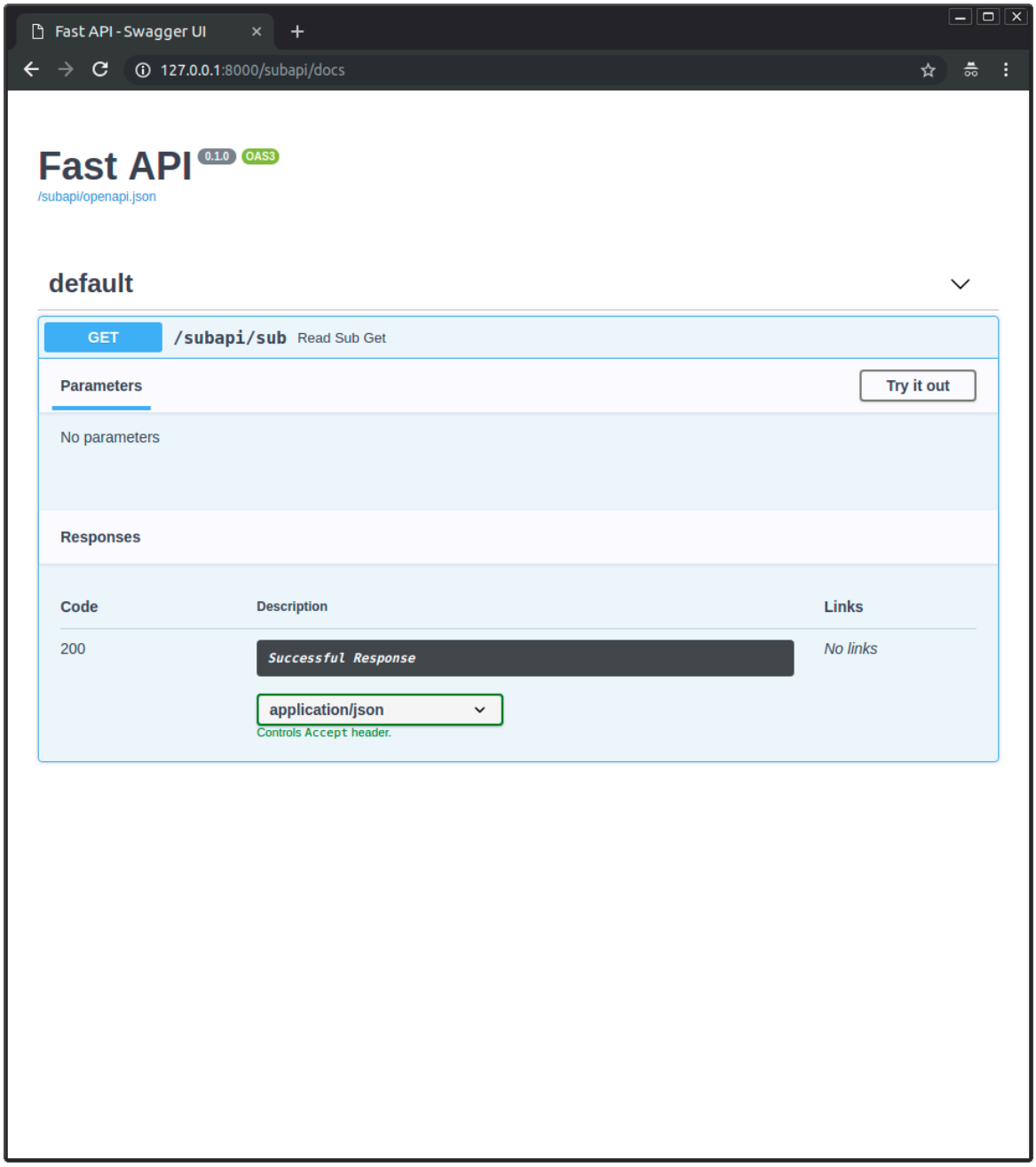
And open the docs at http://127.0.0.1:8000/docs.

You will see the automatic API docs for the main app, including only its own *path operations*:



And then, open the docs for the sub-application, at http://127.0.0.1:8000/subapi/docs.

You will see the automatic API docs for the sub-application, including only its own *path operations*, all under the correct sub-path prefix `/subapi`:



If you try interacting with any of the two user interfaces, they will work correctly, because the browser will be able to talk to each specific app or sub-app.

Technical Details: `root_path`

When you mount a sub-application as described above, FastAPI will take care of communicating the mount path for the sub-application using a mechanism from the ASGI specification called a `root_path`.

That way, the sub-application will know to use that path prefix for the docs UI.

And the sub-application could also have its own mounted sub-applications and everything would work correctly, because FastAPI handles all these `root_path`s automatically.

You will learn more about the `root_path` and how to use it explicitly in the section about Behind a Proxy.